
5 Reference points (TOR 5)

Deborah R Hart and Jui-Han Chang

Northeast Fisheries Science Center, Woods Hole MA

Update or redefine status determination criteria (SDC; point estimates or proxies for B_{MSY} , $B_{THRESHOLD}$, F_{MSY} and MSY reference points) and provide estimates of those criteria and their uncertainty, along with a description of the sources of uncertainty. If analytic model-based estimates are unavailable, consider recommending alternative measurable proxies for reference points. Compare estimates of current stock size and fishing mortality to existing, and any redefined, SDCs. Provide stock status based on updated reference points.

5.1 Introduction

Per recruit reference points were used as proxies for F_{MSY} and B_{MSY} in scallop assessments prior to 2010. The per recruit reference point F_{MAX} , the fully recruited fishing mortality rate that generates maximum yield-per-recruit, was used as a proxy for F_{MSY} . The biomass reference point was defined as the product of B_{MAX} (biomass per recruit at $F = F_{MAX}$) and median numbers of recruits. As selectivity has shifted to larger scallops, yield per recruit curves have become increasingly flat, particularly in the Mid-Atlantic, making per-recruit reference points unstable. Additionally, recruitment has been stronger during periods when biomass has been high, suggesting that spawner-recruit relationships should be included. Finally, risk-based reference points are needed to calculate Acceptable Catch Levels/Allowable Biological Catch (ACLs/ABCs) and target fishing mortalities.

The stochastic yield model (SYM) (Hart 2013) was developed to address these issues; it has been used calculate reference points in scallop assessments since 2010 (NEFSC 2010). It uses Monte-Carlo simulations to propagate the uncertainty in per recruit and stock-recruit calculations while calculating yield curves, F_{MSY} and B_{MSY} . Separate SYM models were run; one each for the Mid-Atlantic and Georges Bank, and their results combined to obtain whole-stock reference points. Because of its limited time series and minor contribution to landings and biomass, Gulf of Maine scallops were not included in these calculations. Georges Bank reference points are used as proxy reference points for Gulf of Maine scallops.

5.2 Methods

The SYM model combines per-recruit calculations with stock-recruit relationships in order to estimate yield curves, as discussed in Beverton and Holt (1957) and Shepherd (1982). However, the SYM approach treats both the per-recruit and the stock-recruit relationships as being uncertain, and takes this uncertainty into account.

Although the SYM model is separate from CASA, efforts were made to make the two models as compatible as possible. In particular, growth was modeled using stochastic growth matrices based on the most recent period. However, because the SYM model (unlike CASA) uses a stock-recruit relationship, the Georges Bank open and closed areas were combined, so that two SYM models were used, one for Mid-Atlantic and one for Georges Bank.

Uncertainties in the SYM model can be divided into uncertainty in the per recruit models, and that from the stock recruit relationship. Per recruit calculations depend on a number of parameters that each carry a level of uncertainty, including shell height/meat weight parameters, fishery selectivity, cull size and the fraction of discards that survive, the incidental mortality rate, and the natural mortality rate M . Each of these was modeled by specifying a distribution, together with parameters for that distribution, typically a mean and variance. A more detailed discussion of all these parameters can be found in Hart (2013). The form of the distributions used are the same as in Hart (2013), but point estimates of some of the parameters, such as natural mortality, selectivity and shell height/meat weight, have been updated as discussed previously.

Of all the input parameters to the per recruit modeling, by far the most uncertain is natural mortality. Natural mortality was modelled as an inverse gamma distribution (i.e., $1/\gamma$, where γ is a gamma distribution). This makes sense for several reasons. First, in many methods to estimate natural mortality, such as maximum longevity or “clapper” ratios for scallops, the uncertain quantity is in the denominator. Secondly, inverse gamma distributions are skewed to the right, which makes sense for natural mortality. For example, it is possible that the true natural mortality is twice or more than its point estimate, but it cannot be zero or less. For this assessment, the mean M was taken as 0.27 on Georges Bank, consistent with the mean natural mortality in this area used in the CASA models. The Mid-Atlantic model presents more complex issues because of its recent declines in productivity. In order to take this into account, the base Mid-Atlantic SYM model used the mean natural mortality at size from the CASA model in the last five years (2019 – 2023). This resulted in M values ranging from 0.36 for a 50 mm scallop to around 0.56 for large adults. Uncertainties in natural mortalities were set at $\sigma = 0.08$ for both regions (Figure 5.1).

Beverton-Holt stock-recruit curves were fitted to spawning stock biomass B (using meat weight) and recruitment estimates from base CASA model runs

$$R = \frac{sB}{h + B} \quad (1)$$

assuming square-root-normal errors, where s is the expected asymptotic recruitment, and h is the spawning stock biomass where the expected recruitment is half its asymptotic value. Standard errors of the stock-recruit parameters and their correlation were estimated using the delta method.

For this and other recent assessments, CASA estimates recruits as one year olds. These were advanced to age two using the CASA estimated M s by size and year. The SYM model thus starts its per recruit calculations at 50 mm, approximately two years old, and the stock-recruit curve (Equation 1, Figure 5.2) was estimated in terms of two year old recruits. Meat weight was used as the primary surrogate for spawning stock biomass (SSB), although there was some exploration using of gonad weight instead of meat weight.

Recruitment in the Mid-Atlantic has been poor in recent years, likely due to shifts in environmental conditions. Actual recruitment during the last five years in this area has been on average 41% of that predicted by the stock-recruitment relationship. For that reason, recruitment in the base Mid-Atlantic SYM model was reduced by a factor of 0.41 from the prediction of the stock-recruit relationship. In addition, CASA estimated adult natural mortality has been higher in recent years in this region; for this reason, the base SYM run used the mean CASA-estimated natural mortality from the last five years (about 0.56 for adult scallops). Sensitivities were performed without these adjustments.

At each iteration of the simulation model, parameter values were drawn from their corresponding distributions, and per recruit and yield curves were calculated. This was repeated 100,000 times and the results of each iteration were stored. The stock-recruit parameters were simulated as correlated square-root normals.

For each run, equilibrium recruitment at fishing mortality F is given by:

$$R = s - h/b(F) \quad (2)$$

where $b(F)$ is SSB per recruit at fishing mortality F , and s and h are as in equation (1). Total yield $Y(F)$ is therefore:

$$Y(F) = y(F)R = y(F)[s - h/b(F)] \quad (3)$$

where $y(F)$ is yield per recruit at fishing mortality F .

Mean yield curves calculated by this method can be disproportionately influenced by outliers, both in cases where the population collapses at zero fishing and where predicted yields are unrealistically high (Hart 2013). For this reason, we used the 10% trimmed mean of the 100,000 runs to obtain point estimates of yield at each fishing mortality, where the highest and lowest 10% of the runs were removed. The probabilistic F_{MSY} was taken as the fishing mortality that maximizes the trimmed mean yield curve; MSY and B_{MSY} are the trimmed mean yield and SSB curves at F_{MSY} . In some previous assessments, the median was used instead because of instability when using the trimmed mean. This was not observed in this assessment, and it was judged that the trimmed mean is a better measure of expected yield than the median.

5.3 Results

Both regions show evidence for higher recruitment at higher SSB, but this pattern is more clear for Georges Bank (Figure 5.2). Estimates of F_{MSY} are highly uncertain, particularly for the Mid-Atlantic, where over a third of the simulations hit the maximum F of 2 (Figure 5.3). Distributions of MSY and B_{MSY} are somewhat less variable, but still have considerable uncertainty. The trimmed mean yield curve (Figure 5.4) shows that Georges Bank is currently estimated to be much more productive and had a much lower F_{MSY} than the Mid-Atlantic. The reference point estimates for the combined stock are $F_{\text{MSY}} = 0.49$ and $B_{\text{MSY}} = 93282$ mt meats (Table 5.1).

If no adjustments for recent conditions were made in the Mid-Atlantic, so $M = 0.4$ and no adjustment for poor recent recruitment, MSY for the Mid-Atlantic would be around 25000 mt meats. Setting M to its estimated value in the last five years in the Mid-Atlantic but not adjusted recruitment reduces the Mid-Atlantic MSY to about 20000 mt meats. The working group judged that adjusting both M and recruitment to reflect recent conditions is more credible.

Comparing these reference points to CASA estimates gives stock status. Combined biomass in 2023 from the three CASA models is 69956 mt meats. This is below the target biomass of $B_{\text{MSY}} = 93282$ but above $B_{\text{threshold}} = 46641$ mt meats, so the stock is not overfished. Combined fully recruited fishing mortality in 2023 was 0.33. This is below $F_{\text{MSY}} = 0.49$ so overfishing is not occurring. It should be noted, however, that fully recruited fishing mortality in both the Georges Bank Closed and Georges Bank Open area in 2023 was about 0.47, which is above the Georges Bank $F_{\text{MSY}} = 0.36$. Thus, if sea scallops were managed as two separate stocks (Mid-Atlantic and Georges Bank), overfishing would be occurring on Georges Bank.

Table 5.1. Summary of reference points and 2023 biomass and fishing mortality for Georges Bank, Mid-Atlantic and combined. Med. Yield_{0.64} and combined.

Region	MSY	F_{MSY}	B_{MSY}	$B_{\text{threshold}}$	B_{2023}	F_{2023}
GB	22706	0.36	83414		49400	0.47
MA	7941	1.56	15909		20556	0.06
Combined	28402	0.49	93282	41707	69956	0.33

Figures

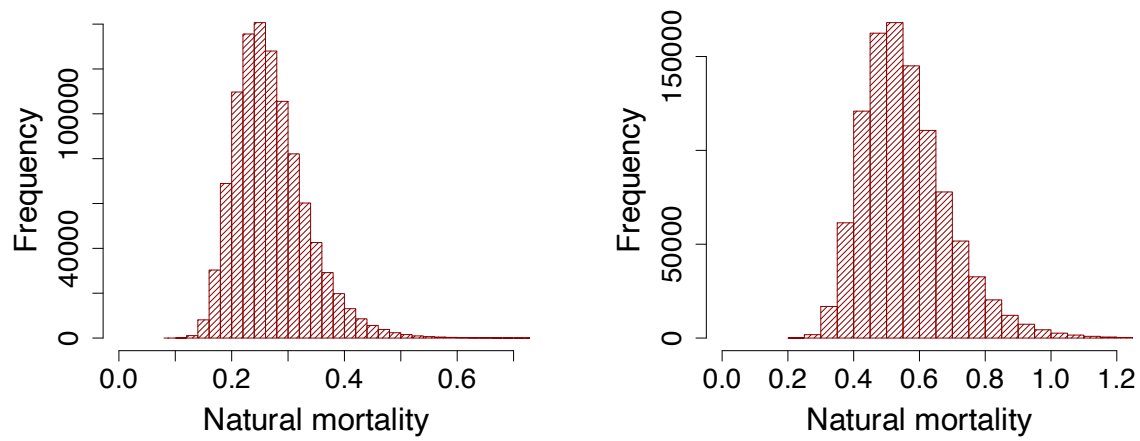


Figure 5.1. Distribution of the natural mortality parameters used in the SYM model for Georges Bank (left) and Mid-Atlantic adults (right)

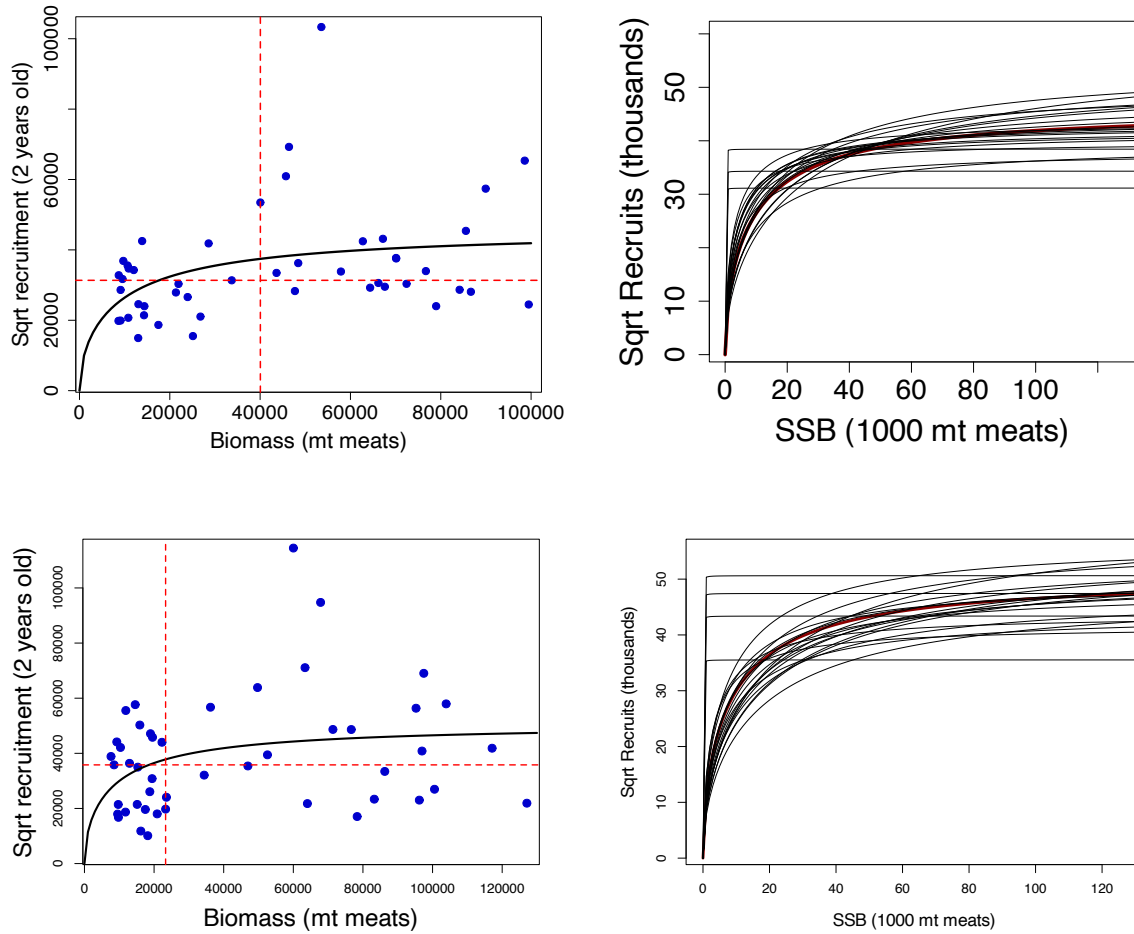


Figure 5.2. Stock-recruit plots for Georges Bank (above left), and Mid-Atlantic (below left), with best fits to the data (black line), and lines indicating the median biomass and recruitment. To the right of these plots are 25 example stock-recruit fits (thin black lines) from the SYM model.

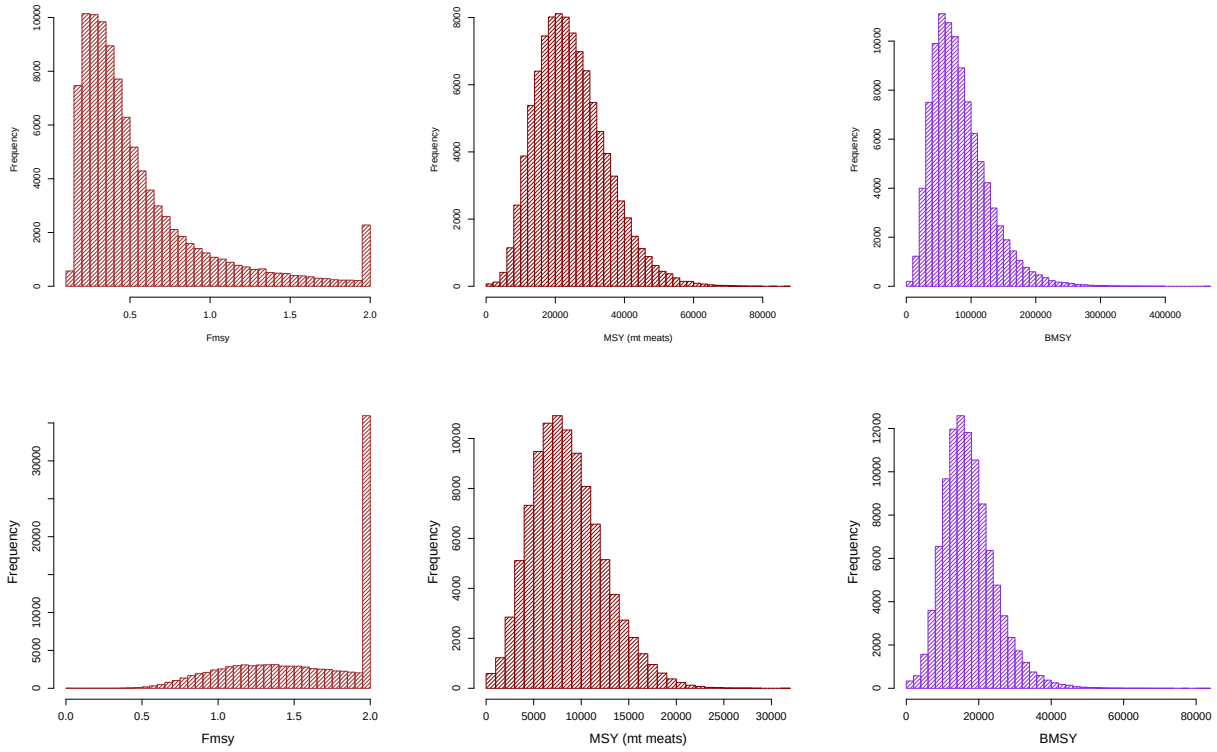


Figure 5.3. Distributions of F_{MSY} , MSY and B_{MSY} from the SYM model for Georges Bank (above) and the Mid-Atlantic (below).

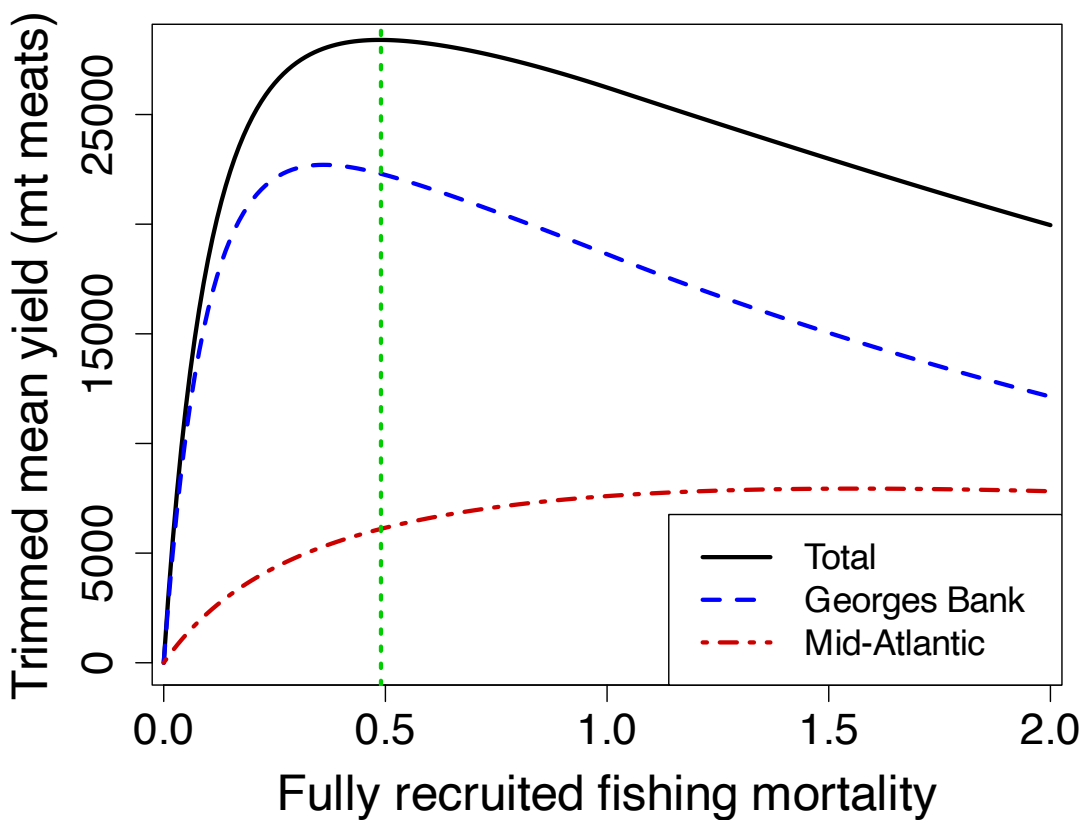


Figure 5.4. Trimmed mean yield curves for the Mid-Atlantic, Georges Bank and combined. The green vertical line indicates the combined $F_{MSY} = 0.49$

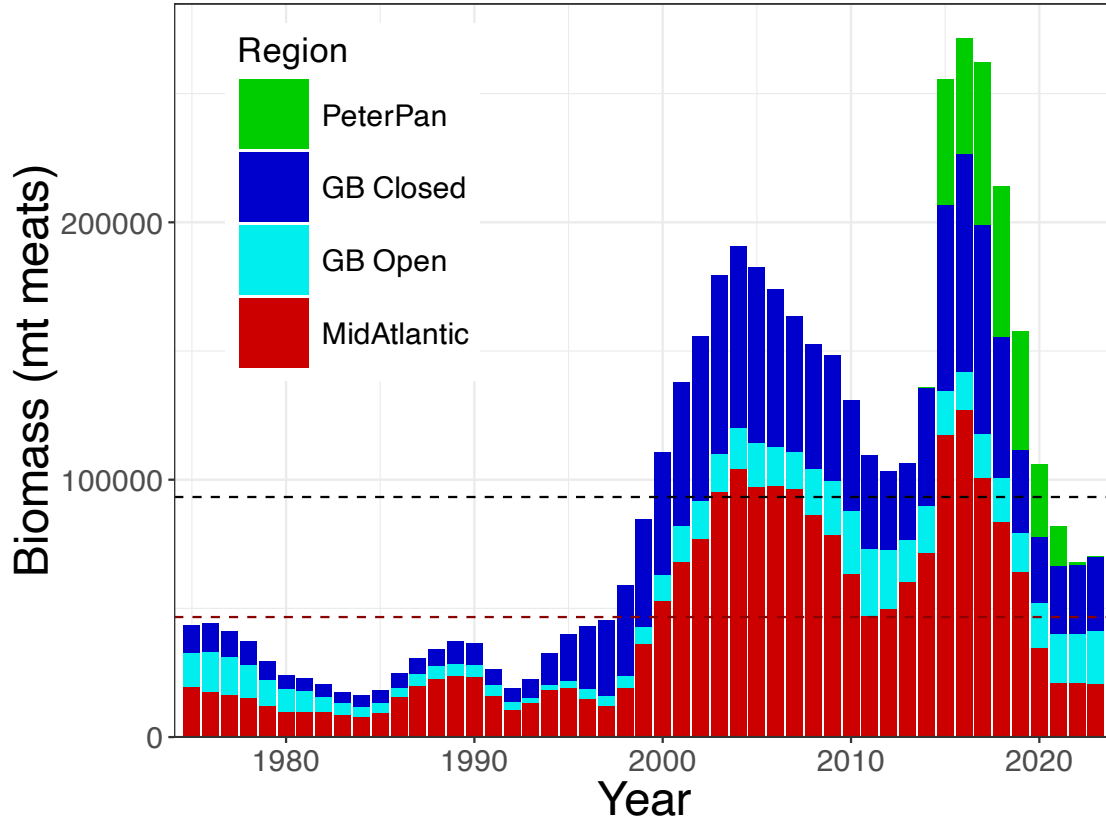


Figure 5.5. Stacked plot showing CAsA estimated biomass over time for the Mid-Atlantic, Georges Bank Closed and Open, and Habcam estimates for the “Peter Pan” southeastern Nantucket Lightship area. The black dashed line indicates the target biomass B_{MSY} and the red dashed line the threshold biomass $\frac{1}{2}B_{MSY}$.

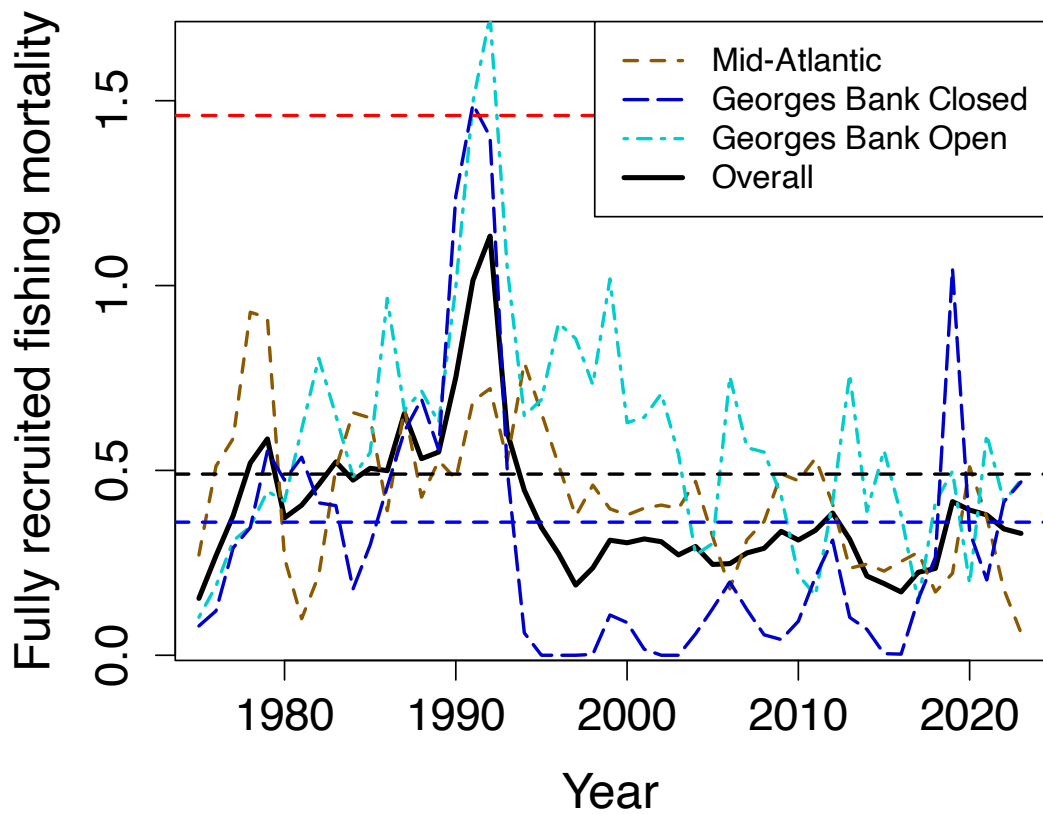


Figure 5.6. Estimated fishing mortality in Georges Bank Open, Closed and in the Mid-Atlantic, compared with F reference points for the Mid-Atlantic (red horizontal dashed line), Georges Bank (blue horizontal dashed line), and overall (black horizontal dashed line). Note that these reference points are only applicable for the last five years.